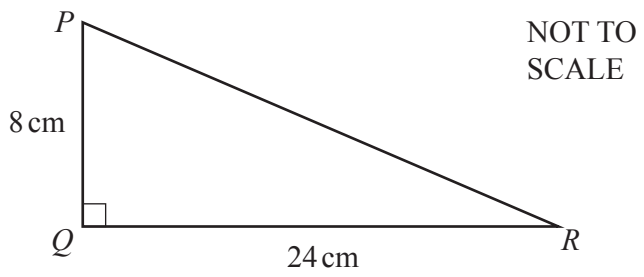
0580_s23_qp_41_Q4

4 **(a)**



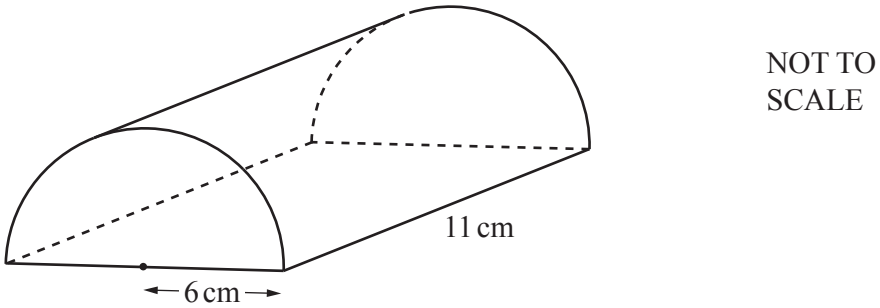
(i) Calculate the area of triangle *PQR*.

..... cm^2 [2]

(ii) Calculate angle *PRQ*.

Angle *PRQ* = ... [2]

(b)



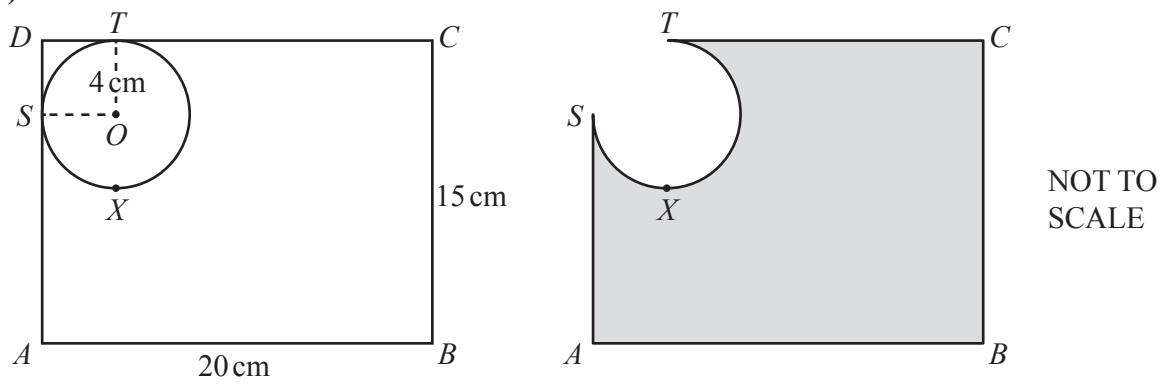
The diagram shows a half-cylinder of radius 6 cm and length 11 cm.

Calculate the volume of the half-cylinder.

..... cm^3 [2]

0580_s23_qp_41_Q4

(c)



(i) *ABCD* is a rectangle with $AB = 20$ cm and $BC = 15$ cm.
S, *X* and *T* are points on a circle centre *O*, such that *DSA* and *DTC* are tangents to the circle.
The radius of the circle is 4 cm and *TX* is a diameter of the circle.
The shape *DSXT* is removed from the corner of the rectangle, leaving the shaded shape shown in the second diagram.

Calculate the area of the shaded shape.

..... cm^2 [5]

(ii) Calculate the perimeter of the shaded shape.

..... cm [3]